Note: TIMx is used as a general term to refer to a timer (for example, TIM1).

Refer to *Section 5.3.13: I/O port characteristics* for details on the input/output alternate function characteristics (output compare, input capture, external clock, PWM output).

Table 60. TIMx characteristics

Symbol	Parameter	Conditions	Min(1)	Max(1)	Unit
$t_{res(TIM)}$	Timer resolution time	-	1	-	$t_{TIMxCLK}$
		$f_{TIMxCLK}$ = 48 MHz	20.833	-	ns
f_{EXT}	Timer external clock frequency on CH1 to CH4	-	0	$f_{TIMxCLK}/4$	MHz
Res_{TIM}	Timer resolution	TIMx other than TIM2	-	16	bit
		TIM2	-	32	
$t_{COUNTER}$	Counter clock period	TIMx other than TIM2	1	2^{16}	$t_{TIMxCLK}$
		TIM2	1	2^{32}	

1. Specified by design. Not tested in production.

Table 61. IWDG min/max timeout period at 32 kHz LSI clock

Prescaler divider	PR[2:0] bits	Min timeout(1) RL[11:0]= 0x000	Max timeout(1) RL[11:0] = 0xFFF	Unit
/4	0	0.125	512	ms
/8	1	0.250	1024	
/16	2	0.500	2048	
/32	3	1.0	4096	
/64	4	2.0	8192	
/128	5	4.0	16384	
/256	6 or 7	8.0	32768	

1. The exact timings further depend on the phase of the APB interface clock versus the LSI clock, which causes an uncertainty of one RC period.

5.3.19 Characteristics of communication interfaces

I^2C-bus interface characteristics

The I^2C-bus interface meets timing requirements of the I^2C-bus specification and user manual rev. 03 for:

- Standard-mode (Sm): with a bit rate up to 100 kbit/s
- Fast-mode (Fm): with a bit rate up to 400 kbit/s
- Fast-mode Plus (Fm+): with a bit rate up to 1 Mbit/s.

The timings are specified by design as long as the I2C peripheral is properly configured (refer to the reference manual RM0490) and when the I2CCLK frequency is greater than the minimum shown in the following table.